

Jump-Model Regimes and Two-Asset Allocation

Data and return-feature construction

- Monthly sample: **1970-01-31 to 2024-11-30**
- Series: equity returns, aggregate bond returns, risk-free return, and macro predictors
- Asset excess return: $r_t^e = r_t - r_t^f$
- Exponentially weighted mean with halflife h :

$$\mu_{t,h} = \frac{\sum_{j \geq 0} 2^{-j/h} r_{t-j}^e}{\sum_{j \geq 0} 2^{-j/h}}$$

- Downside deviation:

$$DD_{t,h} = \sqrt{\frac{\sum_{j \geq 0} 2^{-j/h} \min(r_{t-j}^e, 0)^2}{\sum_{j \geq 0} 2^{-j/h}}}$$

- Log downside deviation:

$$\log DD_{t,h} = \log(DD_{t,h} + \varepsilon)$$

- Equity risk-adjusted feature:

$$\text{Sortino}_{t,h} = \frac{\mu_{t,h}}{DD_{t,h} + \varepsilon}$$

- Bond risk-adjusted feature:

$$\text{Sharpe}_{t,h} = \frac{\mu_{t,h}}{\sigma_{t,h}}$$

Feature families and macro block

- Two monthly feature families with different halflife sets
- Halflives give smooth recursive features with short- and medium-horizon memory
- Shared macro block:

$$(\Delta y_t^{(2Y)}, \text{slope}_t, \Delta \text{slope}_t, \text{VIXtrend}_t, \rho_{t,12m}^{EQ,FI})$$

- These are:
 - 2Y yield change
 - 10Y–2Y slope level
 - 10Y–2Y slope change
 - VIX trend
 - rolling equity–bond correlation

For each asset and month, the jump model observes a transformed feature vector z_t built from the corresponding return-feature family and the shared macro block.

Jump-model regime identification

For each asset, the two-state jump model solves

$$\min_{\{s_t\}, \{\mu_k\}} \sum_{t=1}^T \|z_t - \mu_{s_t}\|^2 + \lambda \sum_{t=2}^T \mathbf{1}(s_t \neq s_{t-1}), \quad K = 2.$$

- $s_t \in \{1, 2\}$: latent state at month t
- μ_k : feature-space center of state k
- λ : persistence penalty; larger λ gives stickier state paths

Raw states are mapped to bull and bear by cumulative excess return:

$$\text{bull} = \arg \max_k \sum_{t: s_t=k} r_t^e, \quad \text{bear} = \arg \min_k \sum_{t: s_t=k} r_t^e.$$

Final choice

$$\lambda^{FI} = 3, \quad \lambda^{EQ} = 10$$

Bond: best annualized Sharpe among non-degenerate candidates with genuine switching.

Equity: best annualized Sharpe and strongest next-state performance before collapse.

Lambda - bond

λ	balance	sw.	$P(00, 01, 10, 11)$	run	bull / bear ex.	next acc.	bal. acc.	Sharpe	invest
0-0.03	58.4/41.6	117	[.788, .212, .153, .847]	5.585	0.0091/ — 0.0098	0.829	0.819	0.704	62.1%
0.1	57.8/42.2	113	[.798, .202, .150, .850]	5.781	0.0091/ — 0.0096	0.836	0.827	0.727	61.4%
0.3	58.0/42.0	107	[.808, .192, .141, .859]	6.102	0.0090/ — 0.0096	0.843	0.834	0.739	61.4%
1	58.4/41.6	89	[.839, .161, .117, .883]	7.322	0.0087/ — 0.0093	0.874	0.866	0.794	61.9%
3	59.9/40.1	59	[.890, .110, .076, .924]	10.983	0.0077/ — 0.0086	0.918	0.911	0.859	64.2%
10	73.6/26.4	24	[.931, .069, .025, .975]	26.360	0.0052/ — 0.0100	0.968	0.954	0.855	77.6%
30	75.7/24.3	3	[.994, .006, .004, .996]	164.750	0.0033/ — 0.0054	0.997	0.995	0.760	80.4%
100	71.5/28.5	2	[.995, .005, .002, .998]	219.667	0.0032/ — 0.0039	0.997	0.995	0.760	80.4%

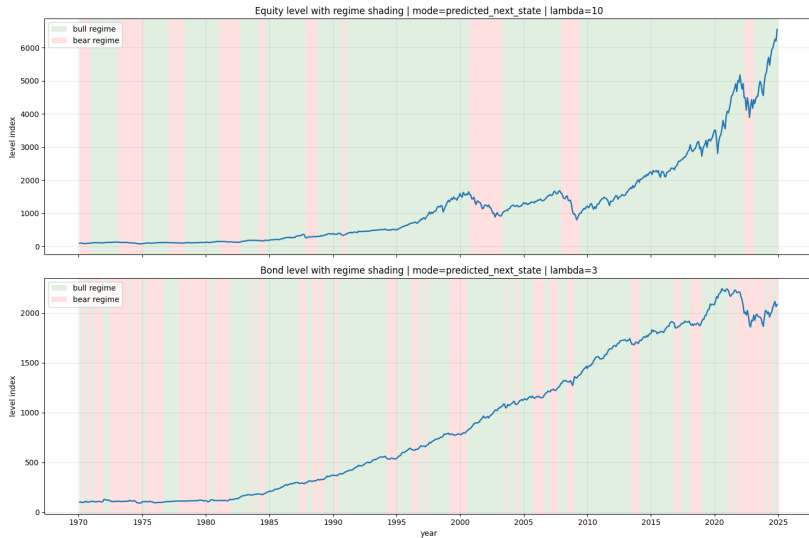
Chosen: $\lambda = 3$. Larger values start collapsing the two-state process; smaller values switch too frequently.

Lambda screening - equity

λ	balance	sw.	$P(00, 01, 10, 11)$	run	bull / bear ex.	next acc.	bal. acc.	Sharpe	invest
0-0.03	55.8/44.2	131	[.773, .227, .177, .823]	4.992	0.0214/ — 0.0186	0.801	0.798	0.278	55.8%
0.1	57.4/42.6	119	[.786, .214, .156, .844]	5.492	0.0208/ — 0.0192	0.819	0.815	0.345	57.3%
0.3	58.1/41.9	111	[.797, .203, .144, .856]	5.884	0.0202/ — 0.0192	0.831	0.827	0.363	58.1%
1	66.6/33.4	69	[.841, .159, .078, .922]	9.414	0.0168/ — 0.0223	0.895	0.883	0.481	66.6%
3	69.0/31.0	49	[.877, .123, .053, .947]	13.180	0.0151/ — 0.0216	0.926	0.913	0.576	69.0%
10	77.8/22.2	19	[.932, .068, .018, .982]	32.950	0.0111/ — 0.0220	0.971	0.959	0.643	77.8%
30	96.4/3.6	4	[.917, .083, .003, .997]	131.800	0.0033/0.0153	0.994	0.957	0.278	96.4%
100	99.4/0.6	2	[.750, .250, .002, .998]	219.667	0.0036/0.0281	0.997	0.874	0.296	99.4%

Chosen: $\lambda = 10$. It maximizes Sharpe before the state process becomes nearly one-sided.

Chosen state paths



Forecasting the final regime states

Final carried-forward targets:

$$\lambda^{FI} = 3, \quad \lambda^{EQ} = 10$$

$$y_{t+1} \in \{0, 1\}, \quad y_{t+1} = 1 \iff \text{bull at } t + 1$$

Rolling design

- training window: 11 years
- validation window: 5 years
- update frequency: every 6 months
- within each update:



- classifier output:
 $p_t = \Pr(y_{t+1} = 1 \mid X_t)$

- trading rule: $\hat{y}_{t+1} = \mathbf{1}\{p_t \geq 0.5\}$

After each 6-month test block, the whole window moves forward by 6 months and the procedure repeats.

Final predictor sets Equity (13):

(logdd_{h1}, logdd_{h4}, mean_{h1}, mean_{h2}, mean_{h4},
sortino_{h1}, sortino_{h2}, sortino_{h4},
ratechg_{2y}, slope, Δslope, vixtrend, $\rho_{12m}^{EQ,FI}$)

Bonds (11):

(mean_{h1}, mean_{h2}, mean_{h4}, sharpe_{h1}, sharpe_{h2}, sharpe_{h4},
ratechg_{2y}, slope, Δslope, vixtrend, $\rho_{12m}^{EQ,FI}$)

Final predictor sets and frozen specification

Model Selection and Configuration:

- **Shallow Trees:** XGBoost grid restricted to $max_depth \in \{2, 3\}$ to prevent overfitting macro noise.
- **Selection Rule:** Hyperparameters optimized via **Validation Sharpe Ratio** (0/1 strategy) rather than raw accuracy.

Equity Predictors (13):

$(\logdd_{h1}, \logdd_{h4}, \text{mean}_{h1}, \text{mean}_{h2}, \text{mean}_{h4},$
 $\text{sortino}_{h1}, \text{sortino}_{h2}, \text{sortino}_{h4},$
 $\text{ratechg}_{2y}, \text{slope}, \Delta\text{slope}, \text{vixtrend}, \rho_{12m}^{EQ, FI})$

Bond Predictors (11):

$(\text{mean}_{h1}, \text{mean}_{h2}, \text{mean}_{h4}, \text{sharpe}_{h1}, \text{sharpe}_{h2}, \text{sharpe}_{h4},$
 $\text{ratechg}_{2y}, \text{slope}, \Delta\text{slope}, \text{vixtrend}, \rho_{12m}^{EQ, FI})$

Final production specification: XGBoost for Bond $\lambda = 3$ and Equity $\lambda = 10$.

Final forecasting results

asset	model	next acc.	bal. acc.	mean ex.	vol	ann. Sharpe
bond	XGBoost	0.8249	0.7788	0.001902	0.009129	0.7218
bond	carry	0.9181	0.9109	0.002559	0.010320	0.8589
equity	XGBoost	0.8777	0.6598	0.005965	0.039481	0.5233
equity	carry	0.9711	0.9592	0.006484	0.034954	0.6426

Differences versus carry benchmark:

$$\Delta \text{Sharpe}^{FI} = -0.1371, \quad \Delta \text{mean ex.}^{FI} = -0.000657$$

$$\Delta \text{Sharpe}^{EQ} = -0.1193, \quad \Delta \text{mean ex.}^{EQ} = -0.000520$$

Takeaway

The carry-forward benchmark remained stronger than the supervised XGBoost forecast for both assets.

correct

Two-asset allocation setup

Common evaluation sample:

1992-01-31 to 2024-10-31

Portfolio net growth:

$$1 + r_{t+1}^{P,\text{net}} = (1 - \text{TC}_t)(1 + r_{t+1}^{P,\text{gross}})$$
$$r_{t+1}^{P,\text{excess}} = r_{t+1}^{P,\text{net}} - r_{t+1}^f$$

Turnover and transaction cost:

$$\text{TO}_t = |w_t^E - \tilde{w}_t^E| + |w_t^B - \tilde{w}_t^B|$$
$$\text{TC}_t = c \cdot \text{TO}_t, \quad c = 0.0005$$

Risk and mean inputs:

Σ_t from exponentially weighted covariance of equity/bond excess returns, halflife 12

$$\mu_t^{\text{MV}} = \text{EWM}_{60}(r^e)$$

For regime-conditioned strategies, each asset mean input is the trailing 11-year regime-conditional average.

Allocation rules

Static benchmark

$$w_t = (0.6, 0.4, 0)$$

Plain equal-weight

$$w_t = (0.5, 0.5, 0)$$

Regime equal-weight

$$w_t = \begin{cases} (0.5, 0.5, 0), & (\hat{y}_t^E, \hat{y}_t^B) = (1, 1), \\ (1, 0, 0), & (\hat{y}_t^E, \hat{y}_t^B) = (1, 0), \\ (0, 1, 0), & (\hat{y}_t^E, \hat{y}_t^B) = (0, 1), \\ (0, 0, 1), & (\hat{y}_t^E, \hat{y}_t^B) = (0, 0). \end{cases}$$

Optimizer constraint set:

$$w_t^E \geq 0, \quad w_t^B \geq 0, \quad w_t^E + w_t^B \leq 1, \quad w_t^C = 1 - w_t^E - w_t^B$$

Grid-search objective, step size 0.0025:

$$\max_{w_t} w_t^\top \mu_t - \gamma_{\text{risk}} w_t^\top \Sigma_t w_t - \gamma_{\text{trade}} c \left(|w_t^E - \tilde{w}_t^E| + |w_t^B - \tilde{w}_t^B| \right)$$

Minimum-variance and mean-variance families

Minimum-variance family

Plain MinVar:

$$\mu_t = (0.001, 0.001)$$

Binary regime MinVar:

$$\mu_t^E = \begin{cases} 0.001, & \hat{y}_t^E = 1 \\ 0, & \hat{y}_t^E = 0 \end{cases}, \quad \mu_t^B = \begin{cases} 0.001, & \hat{y}_t^B = 1 \\ 0, & \hat{y}_t^B = 0 \end{cases}$$

Probability-weighted MinVar:

$$\mu_t^E = 0.001 p_t^E, \quad \mu_t^B = 0.001 p_t^B$$

Reported setting:

$$\gamma_{\text{risk}} = 10, \quad \gamma_{\text{trade}} = 1$$

Mean-variance family

Plain MV:

$$\mu_t = \text{EWM}_{60}(r^e)$$

Binary regime MV:

$$\mu_t^E = \begin{cases} \bar{r}_{t,\text{bull}}^E, & \hat{y}_t^E = 1 \\ \bar{r}_{t,\text{bear}}^E, & \hat{y}_t^E = 0 \end{cases}$$

$$\mu_t^B = \begin{cases} \bar{r}_{t,\text{bull}}^B, & \hat{y}_t^B = 1 \\ \bar{r}_{t,\text{bear}}^B, & \hat{y}_t^B = 0 \end{cases}$$

Probability-weighted MV:

$$\mu_t^E = p_t^E \bar{r}_{t,\text{bull}}^E + (1 - p_t^E) \bar{r}_{t,\text{bear}}^E$$

$$\mu_t^B = p_t^B \bar{r}_{t,\text{bull}}^B + (1 - p_t^B) \bar{r}_{t,\text{bear}}^B$$

Reported settings:

$$\gamma_{\text{risk}} = 5 \text{ (plain MV)}, \quad \gamma_{\text{risk}} = 10 \text{ (regime MV)}, \quad \gamma_{\text{trade}} = 1$$

Reported strategy set

- 60/40
- plain equal-weight
- regime-based equal-weight
- plain minimum-variance
- regime-based minimum-variance with binary forecasts
- plain mean-variance
- regime-based mean-variance with binary forecasts

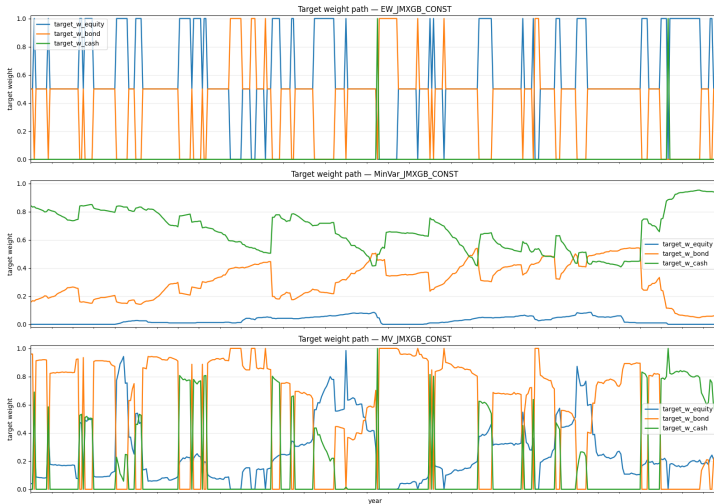
Probability-weighted optimizer variants are robustness checks only.

Allocation results



1. 60_40 is the static benchmark with constant 60% equity and 40% fixed income exposure.
 2. Risk MinVar and plain MV use only historical mean/covariance inputs, with no regime forecast overlay.
 3. CONST strategies use binary next-state forecasts for conditional mean inputs. PROB strategies use forecast probabilities only in the mean inputs.

Target weight paths



These panels show how each strategy reallocates across equity, fixed income, and cash over time. They help separate persistent reallocation behavior from one-off episodic timing gains.

Feature-space coupled-state construction

Keep the existing single-asset JM feature blocks and define summary scores

$$S_t^E = \frac{1}{8} \sum_{j=1}^8 \tilde{z}_{t,j}^E, \quad S_t^B = \frac{1}{6} \sum_{j=1}^6 \tilde{z}_{t,j}^B,$$

where all JM inputs are z-scored and equity downside terms enter with flipped sign, so larger score means better conditions.

Define coupled feature-space states

$$J_t^{BB} = \mathbf{1}\{S_t^E < 0, S_t^B < 0\}, \quad J_t^{GG} = \mathbf{1}\{S_t^E > 0, S_t^B > 0\},$$
$$J_t^{OPP} = \mathbf{1}\{\text{sign}(S_t^E) \neq \text{sign}(S_t^B)\}.$$

Empirical frequencies:

$$\Pr(J_t^{BB} = 1) = 0.2595, \quad \Pr(J_t^{GG} = 1) = 0.3035, \quad \Pr(J_t^{OPP} = 1) = 0.4370.$$

Conditional probabilities in same point of time.

Representative conditioning block

Retained macro-financial representatives:

$x_t = (\text{raw_DGS1}, \text{derived_belly_slope}, \text{CORESTICKM159SFRBATL}, \text{VIXCLSx}, \text{mktrf}, \text{bond_10y_mom_12m}, \text{NAPMPRI})$.

Interpretation:

- raw_DGS1: short-rate level
- derived_belly_slope: curve-shape proxy
- CORESTICKM159SFRBATL: sticky core inflation
- VIXCLSx: market stress
- mktrf: equity trend
- bond_10y_mom_12m: bond trend
- NAPMPRI: cyclical / inflation-sensitive activity proxy

Each variable enters standardized:

$$z_{t,j} = \frac{x_{t,j} - \bar{x}_j}{s_j}.$$

Screening equations

For each coupled state $Y_t \in \{J_t^{BB}, J_t^{GG}, J_t^{OPP}\}$, run

$$\Pr(Y_t = 1 \mid z_{t,j}) = \Lambda(\alpha_j + \beta_j z_{t,j}), \quad \Lambda(u) = \frac{e^u}{1 + e^u},$$

and the joint screen

$$\Pr(Y_t = 1 \mid z_t) = \Lambda\left(\alpha + \sum_{j=1}^7 \beta_j z_{t,j}\right).$$

Report:

- univariate slope β_j and p-value
- multivariate slope β_j and p-value
- low/high bucket frequencies

Univariate screen gives direction. Multivariate screen gives retained effect after overlap removal.

Joint-bad state: results

Univariate logit: all seven representatives significant.

Main univariate slopes:

$$\beta_{\text{curve}}^{BB} = -1.3378, \quad p < 10^{-6}, \quad \beta_{\text{bond}}^{BB} = -1.2223, \quad p < 10^{-6},$$

$$\beta_{\text{equity}}^{BB} = -0.9143, \quad p < 10^{-6}, \quad \beta_{\text{rate}}^{BB} = 0.8973, \quad p < 10^{-6},$$

$$\beta_{\text{activity}}^{BB} = 0.8722, \quad p < 10^{-6}, \quad \beta_{\pi}^{BB} = 0.6559, \quad p < 10^{-6}.$$

Multivariate retained drivers:

$$\text{mktrf} : \beta = -1.0845, \quad p < 10^{-6},$$

$$\text{bond_10y_mom_12m} : \beta = -0.8871, \quad p = 0.000002,$$

$$\text{derived_belly_slope} : \beta = -0.8244, \quad p = 0.000027,$$

$$\text{NAPMPRI} : \beta = 0.5136, \quad p = 0.000169.$$

Joint bad is primarily weak equity trend, weak bond trend, adverse curve geometry, and weak cyclical backdrop.

Joint-good state: results

Univariate logit: all seven representatives significant.

Main univariate slopes:

$$\begin{aligned}\beta_{\text{equity}}^{GG} &= 0.7770, \quad p < 10^{-6}, & \beta_{\text{bond}}^{GG} &= 0.7413, \quad p < 10^{-6}, \\ \beta_{\text{stress}}^{GG} &= -0.7807, \quad p < 10^{-6}, & \beta_{\pi}^{GG} &= -0.7635, \quad p < 10^{-6}, \\ \beta_{\text{activity}}^{GG} &= -0.5820, \quad p < 10^{-6}, & \beta_{\text{curve}}^{GG} &= 0.5385, \quad p < 10^{-6}.\end{aligned}$$

Multivariate retained drivers:

$$\begin{aligned}\text{VIXCLSx} : \beta &= -1.1808, \quad p < 10^{-6}, \\ \text{mktrf} : \beta &= 1.0195, \quad p < 10^{-6}, \\ \text{bond_10y_mom_12m} : \beta &= 0.9934, \quad p < 10^{-6}, \\ \text{CORESTICKM159SFRBATL} : \beta &= -0.7957, \quad p = 0.0016, \\ \text{NAPMPRI} : \beta &= -0.3326, \quad p = 0.0128.\end{aligned}$$

Joint good is the mirror-image state: strong market trend, low stress, lower inflation pressure, and healthier macro conditions.

Opposition state: results

Univariate significant terms:

$$\text{VIXCLSx} : \beta = 0.3097, \quad p = 0.0002,$$

$$\text{derived_belly_slope} : \beta = 0.2733, \quad p = 0.0006,$$

$$\text{raw_DGS1} : \beta = -0.2340, \quad p = 0.0038.$$

Multivariate screen:

$$\text{VIXCLSx} : \beta = 0.3201, \quad p = 0.0003$$

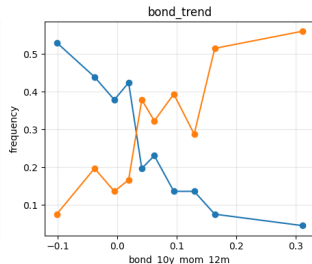
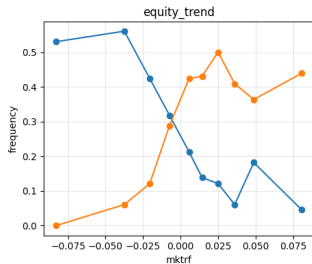
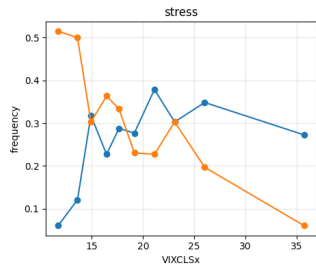
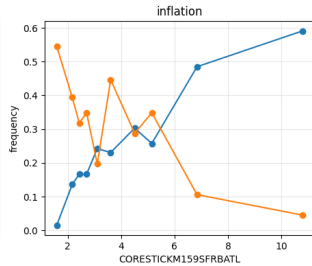
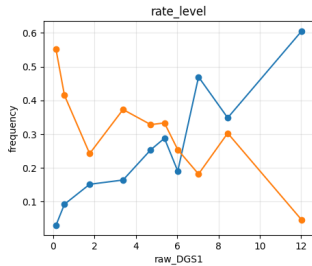
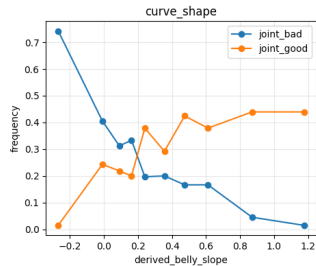
remains the only retained driver, while

$$p_{\text{curve}} = 0.1020, \quad p_{\text{equity}} = 0.1417, \quad p_{\text{activity}} = 0.2581,$$

$$p_{\text{rate}} = 0.2886, \quad p_{\pi} = 0.2988, \quad p_{\text{bond}} = 0.3331.$$

Opposition survives only as a narrow stress-sensitive effect.

Representative signal plots



What the signal plots show

For a retained signal x_t , let $D_1, \dots, D_{10}$ be its deciles. Each panel reports the empirical decile frequencies

$$\Pr(J_t^{BB} = 1 \mid x_t \in D_k) = \frac{1}{|D_k|} \sum_{t \in D_k} J_t^{BB}, \quad \Pr(J_t^{GG} = 1 \mid x_t \in D_k) = \frac{1}{|D_k|} \sum_{t \in D_k} J_t^{GG}, \quad k = 1, \dots, 10.$$

Visible directional patterns:

`raw_DGS1` $\uparrow$, `CORESTICKM159SFRBATL` $\uparrow$, `VIXCLSx` $\uparrow \implies \Pr(J_t^{BB} = 1) \uparrow$, $\Pr(J_t^{GG} = 1) \downarrow$,

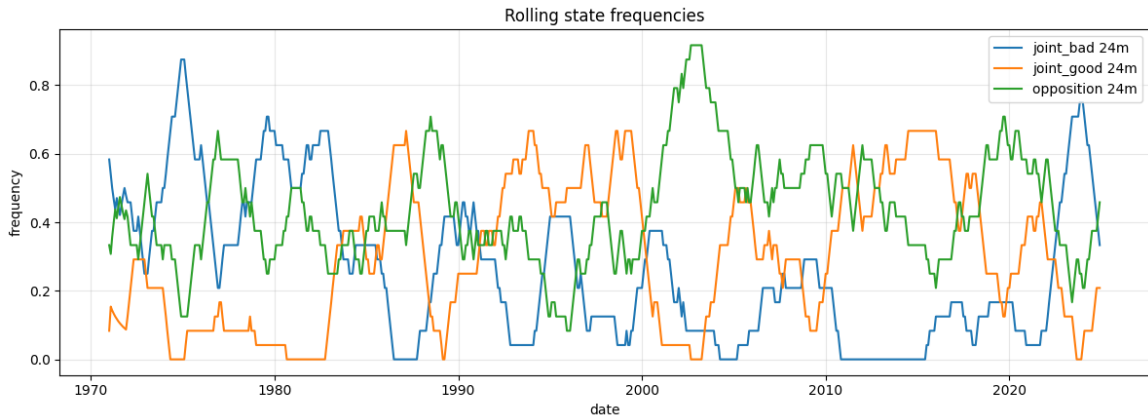
`mktrf` $\uparrow$, `bond_10y_mom_12m` $\uparrow$, `derived_belly_slope` $\uparrow \implies \Pr(J_t^{BB} = 1) \downarrow$, $\Pr(J_t^{GG} = 1) \uparrow$.

So the plots visualize a same-sign good/bad axis:

rates / inflation / stress $\uparrow \implies$ more *BB*, less *GG*,

equity trend / bond trend / curve shape $\uparrow \implies$ less *BB*, more *GG*.

Time variation of coupled feature-space states



Coupling signal block for the direct joint model

Retained raw coupling signals:

$$\mathbf{x}_t = (\text{raw_DGS1}, \text{derived_belly_slope}, \text{CORESTICKM159SFRBATL}, \text{VIXCLSx}, \text{mktrf}, \text{bond_10y_mom_12m}, \text{derived_loa})$$

Model inputs are transformed to a more stationary block

$$\tilde{\mathbf{x}}_t = (\Delta \text{raw_DGS1}, \Delta \text{derived_belly_slope}, \Delta \text{CORESTICKM159SFRBATL}, \Delta \log(\text{VIXCLSx}), \text{mktrf}, \Delta \text{bond_10y_mom_12m})$$

3-state joint jump model

Let

$$\tilde{z}_t = (z_t^{E,\text{own}}, z_t^{B,\text{own}}, z_t^{\text{shared}}, \tilde{x}_t)$$

collect:

- equity-specific JM inputs,
- bond-specific JM inputs,
- shared macro block,
- transformed coupling signals.

The direct joint model solves

$$\min_{\{s_t\}, \{\mu_k\}} \sum_{t=1}^T \|\tilde{z}_t - \mu_{s_t}\|^2 + \lambda \sum_{t=2}^T \mathbf{1}(s_t \neq s_{t-1}), \quad K = 3.$$

Here

$$s_t \in \{1, 2, 3\}$$

is a joint monthly state, estimated directly from the augmented two-asset feature vector.

Time-varying mode-penalty extension

Keep the same 3-state joint structure

$$\mathcal{S} = \{\text{cash_defensive}, \text{bond_defensive}, \text{risk_on}\}.$$

At month t , let

$$m_t = (\text{equity trend}_t, \text{bond trend}_t, \text{stress}_t, \text{rate level}_t)$$

be the focused macro signal block.

A multinomial classifier produces state probabilities

$$\pi_t(k) = \Pr(s_t = k \mid m_t), \quad k \in \mathcal{S}.$$

These are converted into time-varying mode penalties

$$\phi_t(k) = -a \log \pi_t(k), \quad a = 2.$$

Not yet implemented the search for a . Let z_t denote the scaled joint feature vector at month t , and let μ_k be the fitted prototype of state k . The per-period feature-space loss is

$$\ell_t(k) = \|z_t - \mu_k\|_2^2, \quad k \in \mathcal{S}.$$

The decoded state path solves

$$\min_{s_1, \dots, s_T \in \mathcal{S}} \sum_{t=1}^T \left[\ell_t(s_t) + \phi_t(s_t) \right] + \lambda \sum_{t=2}^T \mathbf{1}(s_t \neq s_{t-1}), \quad \lambda = 3.$$

Time-varying mode-penalty extension (cont.)

So:

$\ell_t(s_t)$ = feature fit, $\phi_t(s_t)$ = macro-conditioned state penalty, $\lambda \mathbf{1}(s_t \neq s_{t-1})$ = switching penalty.